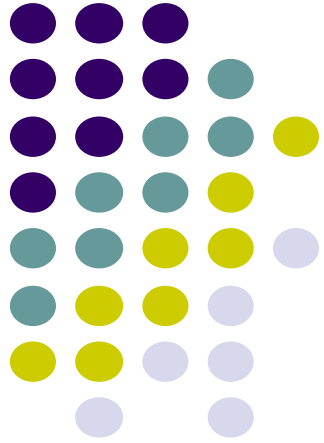
ErSE222: Machine learning in Geoscience

April. 19th, 2026

Tariq Alkhalifah and Omar Saad



RNN

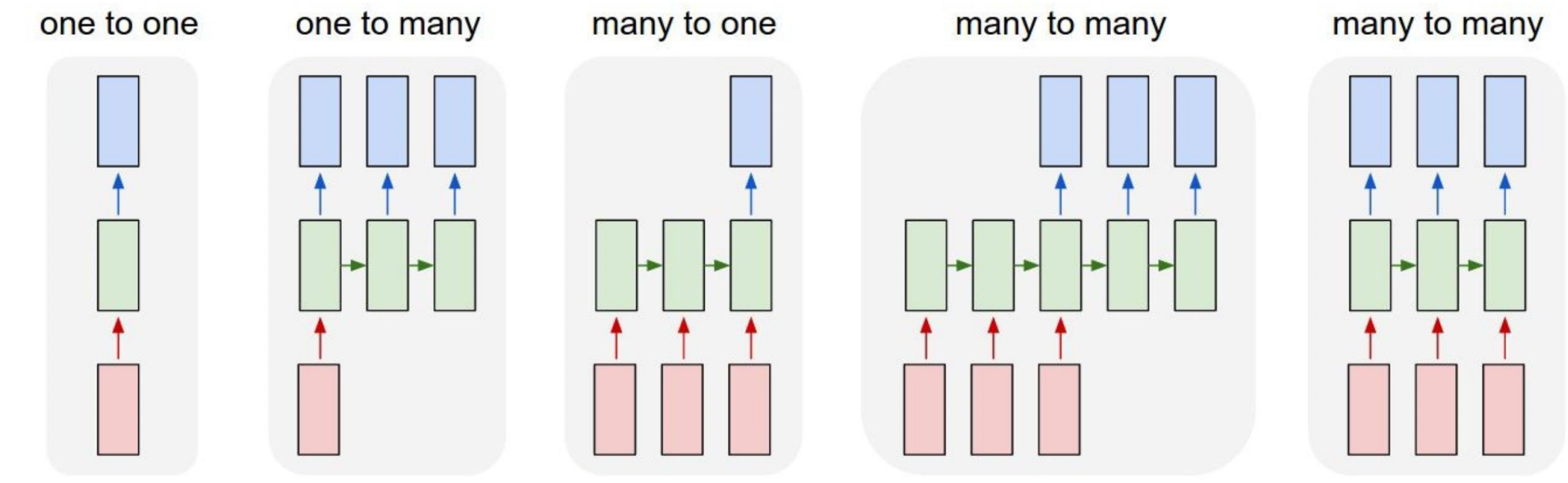


Image Captioning using spatial features

Input: Image **I**
Output: Sequence **y** = $y_1, y_2, ..., y_T$

CNN

$z_{0,0}$	$z_{0,1}$	$z_{0,2}$
$z_{1,0}$	$z_{1,1}$	$z_{1,2}$
$z_{2,0}$	$z_{2,1}$	$z_{2,2}$

Extract spatial features from a pretrained CNN

Features: H x W x D

Xu et al, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Image Captioning using spatial features

Input: Image **I**
Output: Sequence **y** = $y_1, y_2, \ldots, y_T$

Encoder: $h_0 = f_{\mathbf{W}}(\mathbf{z})$
where **z** is spatial CNN features
$f_{\mathbf{W}}(.)$ is an MLP

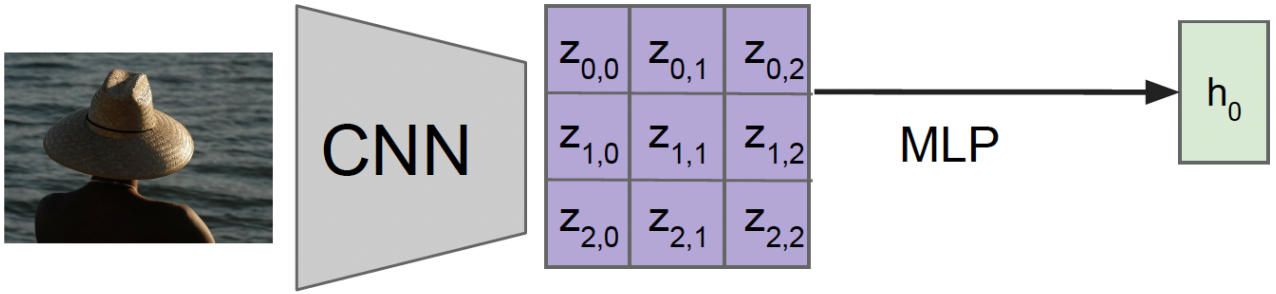


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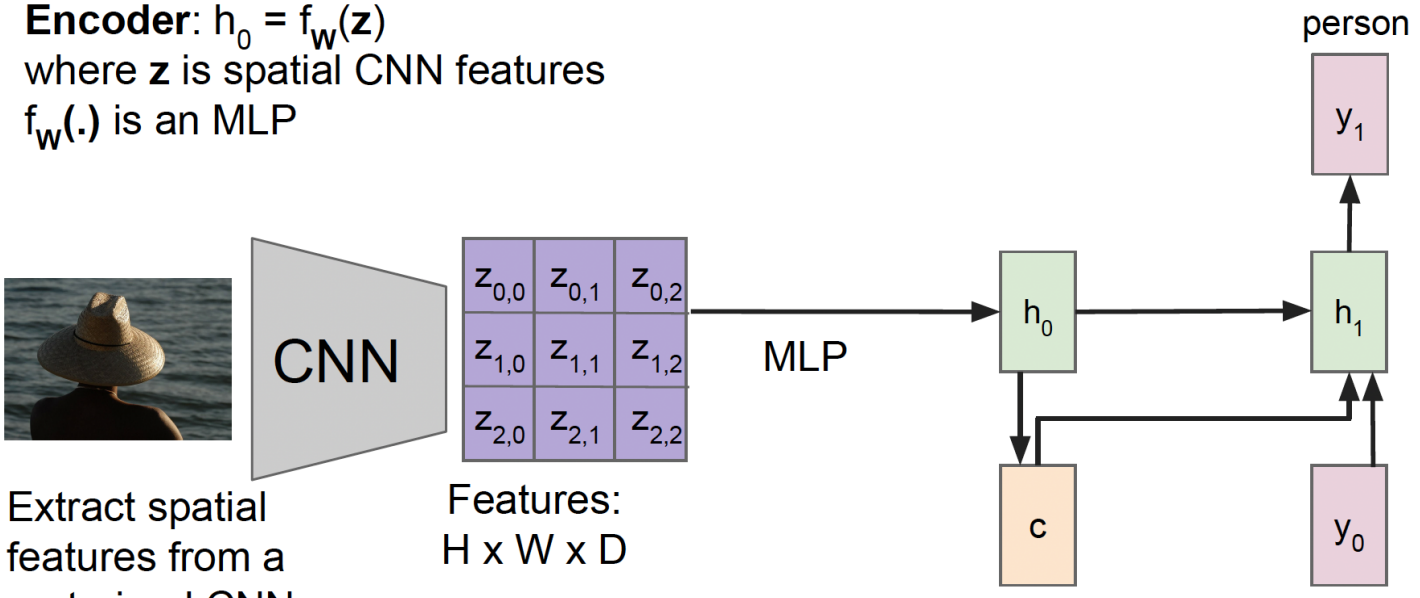
Image Captioning using spatial features

Input: Image **I**
Output: Sequence **y** = $y_1, y_2, \ldots, y_T$

Decoder: $y_t = g_V(y_{t-1}, h_{t-1}, c)$
where context vector c is often $c = h_0$

Encoder: $h_0 = f_W(\mathbf{z})$
where **z** is spatial CNN features
$f_W(.)$ is an MLP



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CNN

$z_{0,0}$ $z_{0,1}$ $z_{0,2}$
$z_{1,0}$ $z_{1,1}$ $z_{1,2}$
$z_{2,0}$ $z_{2,1}$ $z_{2,2}$

Extract spatial features from a pretrained CNN

Features: H x W x D

MLP

person wearing hat [END]

y_1 y_2 y_3 y_4

h_0 h_1 h_2 h_3 h_4

c y_0 y_1 y_2 y_3

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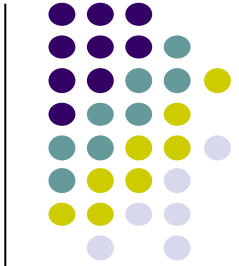
Image Captioning using spatial features



Problem: Input is "bottlenecked" through c

- Model needs to encode everything it wants to say within c

This is a problem if we want to generate really long descriptions? 100s of words long

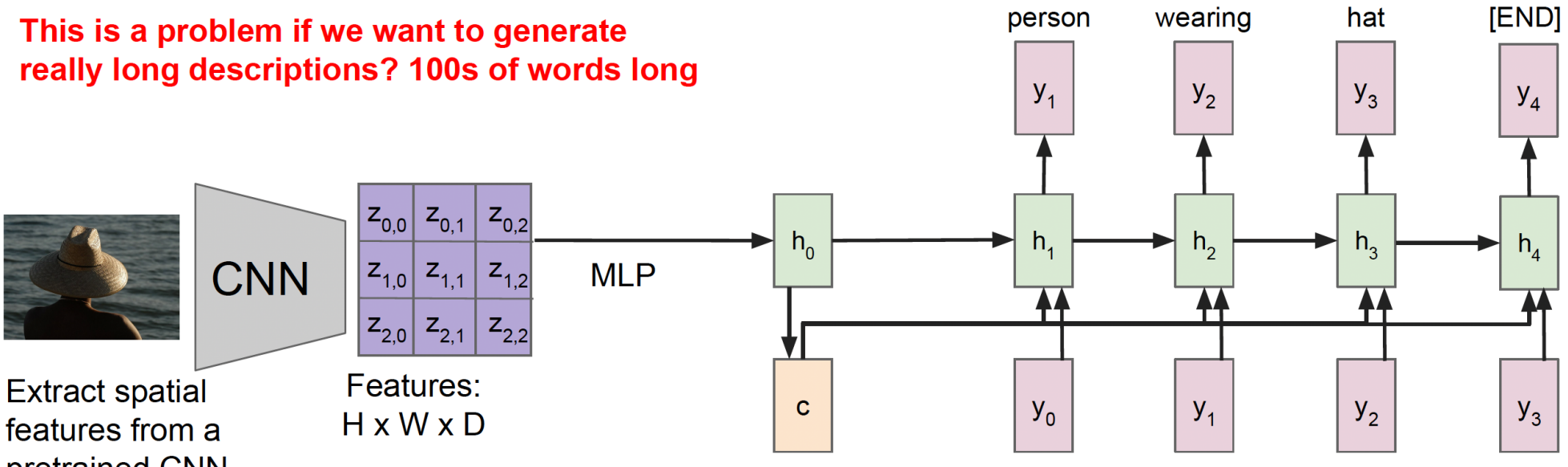


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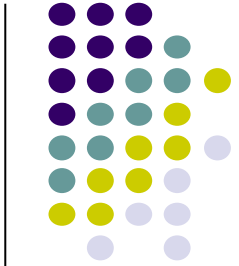
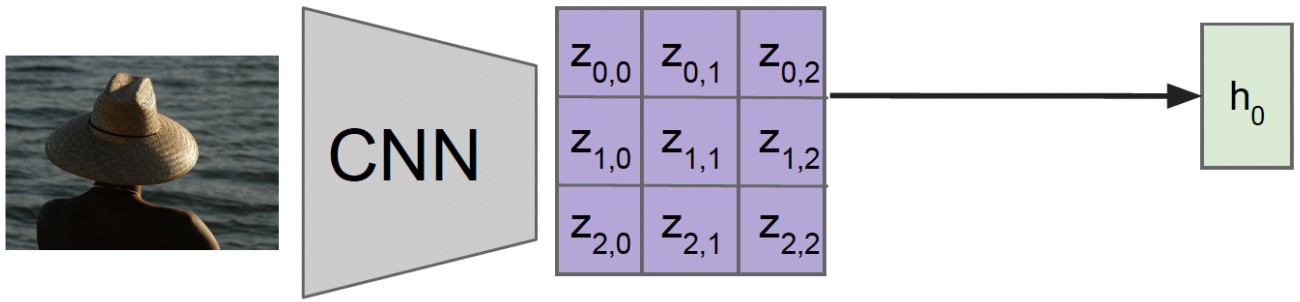
Image Captioning using spatial features



Attention idea: New context vector at every time step.

Each context vector will attend to different image regions



Extract spatial features from a pretrained CNN

Features: H x W x D

Xu et al, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Image Captioning using spatial features and Attention

Compute alignments scores (scalars):

$$e_{t,i,j} = f_{att}(h_{t-1}, z_{i,j})$$

$f_{att}(.)$ is an MLP

Alignment scores: H x W

$e_{1,0,0}$	$e_{1,0,1}$	$e_{1,0,2}$
$e_{1,1,0}$	$e_{1,1,1}$	$e_{1,1,2}$
$e_{1,2,0}$	$e_{1,2,1}$	$e_{1,2,2}$

CNN

Extract spatial features from a pretrained CNN

$z_{0,0}$	$z_{0,1}$	$z_{0,2}$
$z_{1,0}$	$z_{1,1}$	$z_{1,2}$
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Features: H x W x D

h_0

Xu et al, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Image Captioning using spatial features and Attention

CNN

Extract spatial features from a pretrained CNN

Features: H x W x D

$z_{0,0}$ $z_{0,1}$ $z_{0,2}$
$z_{1,0}$ $z_{1,1}$ $z_{1,2}$
$z_{2,0}$ $z_{2,1}$ $z_{2,2}$

h_0

Alignment scores: H x W

$e_{1,0,0}$ $e_{1,0,1}$ $e_{1,0,2}$
$e_{1,1,0}$ $e_{1,1,1}$ $e_{1,1,2}$
$e_{1,2,0}$ $e_{1,2,1}$ $e_{1,2,2}$

Attention: H x W

$a_{1,0,0}$ $a_{1,0,1}$ $a_{1,0,2}$
$a_{1,1,0}$ $a_{1,1,1}$ $a_{1,1,2}$
$a_{1,2,0}$ $a_{1,2,1}$ $a_{1,2,2}$

Compute alignments scores (scalars):

$$e_{t,i,j} = f_{att}(h_{t-1}, z_{i,j})$$

$f_{att}(.)$ is an MLP

Normalize to get attention weights:

$$a_{t,:,:} = softmax(e_{t,:,:})$$

$0 < a_{t,i,j} < 1$, attention values sum to 1

Xu et al, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Image Captioning using spatial features and **Attention**

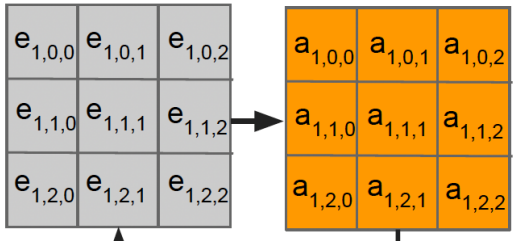
Compute alignments scores (scalars):

$e_{t,i,j} = f_{att}(h_{t-1}, z_{i,j})$

$f_{att}(.)$ is an MLP

Alignment scores: H x W

Attention: H x W



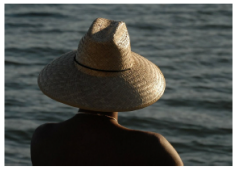
Normalize to get attention weights:

$a_{t,:,:} = softmax(e_{t,:,:})$

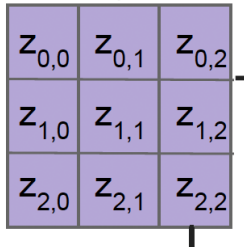
$0 < a_{t,i,j} < 1$,
attention values sum to 1

Compute context vector:

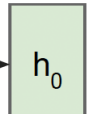
$$c_t = \sum_{ij} a_{t,i,j} z_{t,i,j}$$

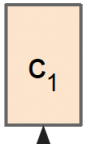


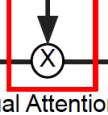
CNN



Features: H x W x D



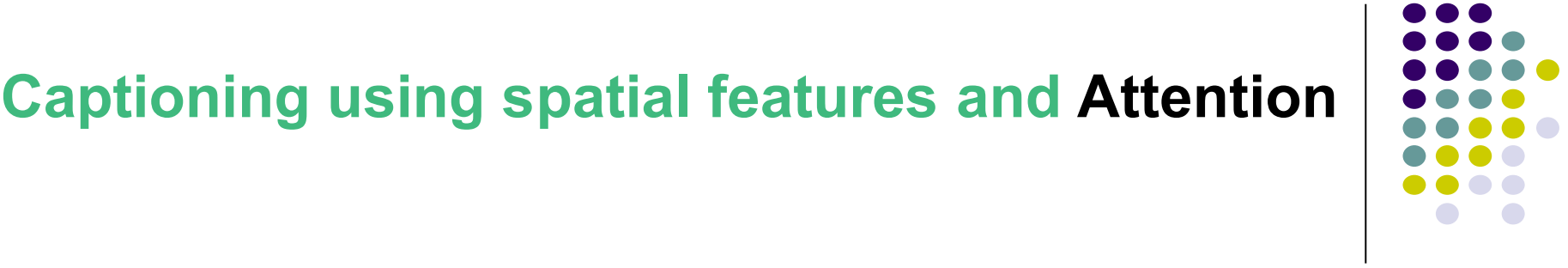




Extract spatial features from a pretrained CNN

Xu et al, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Image Captioning using spatial features and Attention



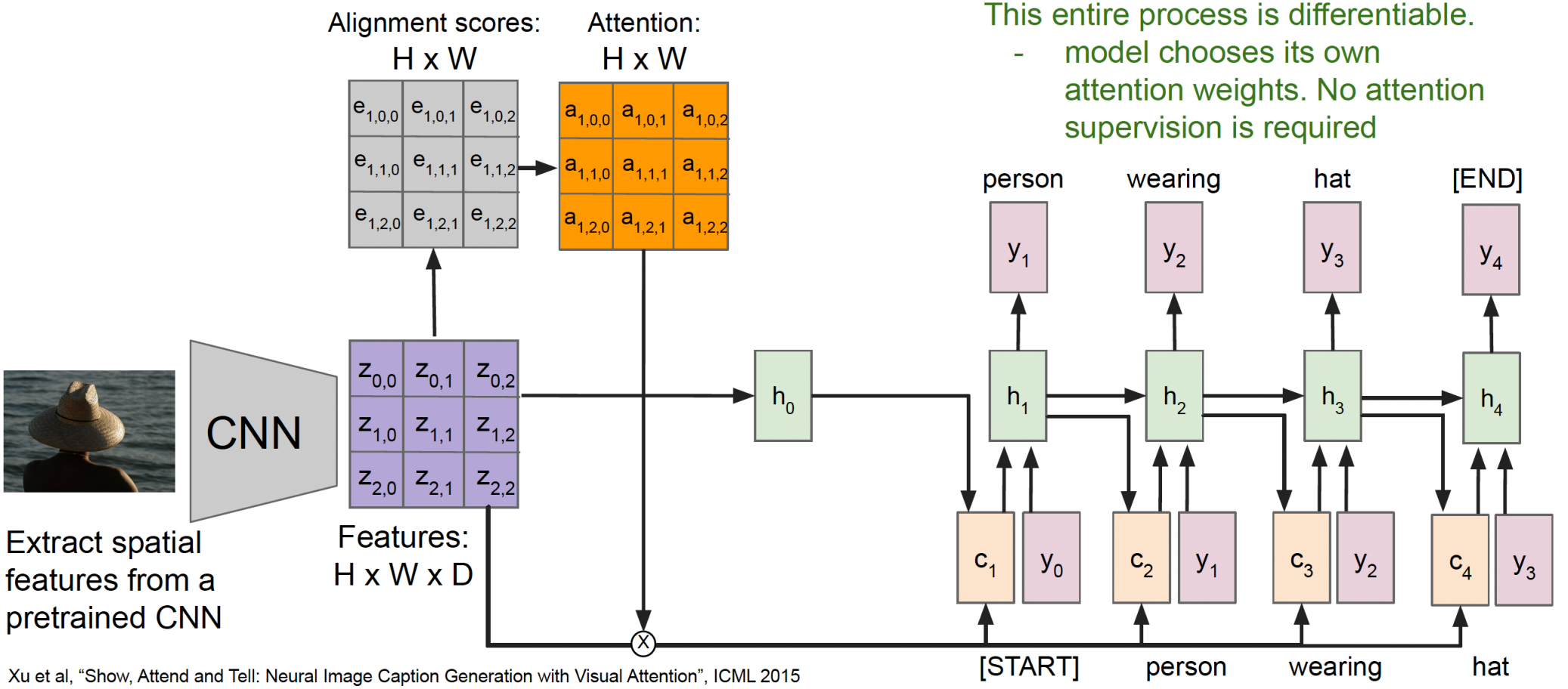


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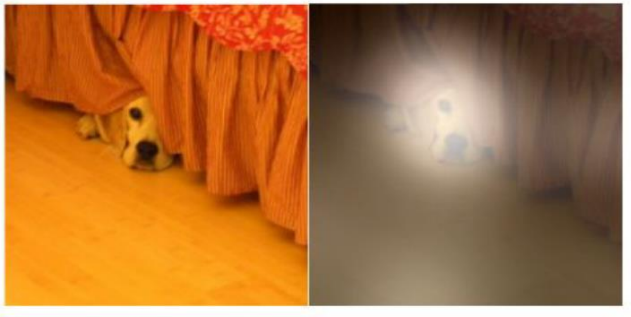
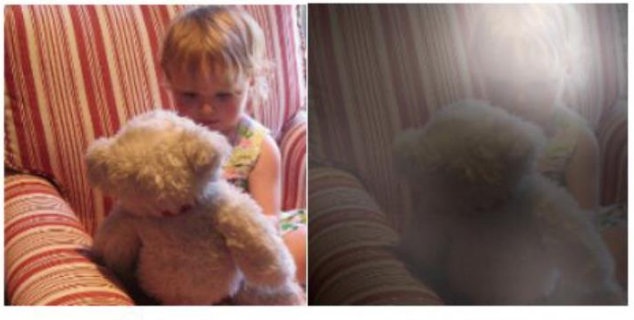
Image Captioning using spatial features and Attention

A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.

A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.

A group of people sitting on a boat in the water.

A giraffe standing in a forest with trees in the background.

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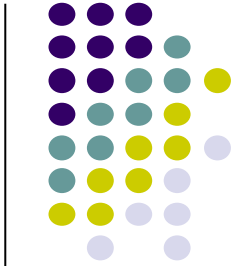
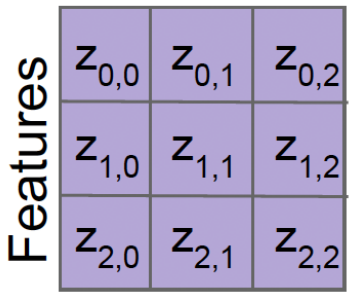
General attention layer



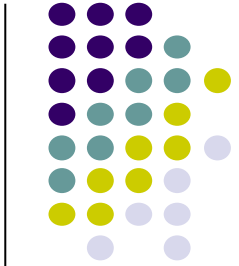


h

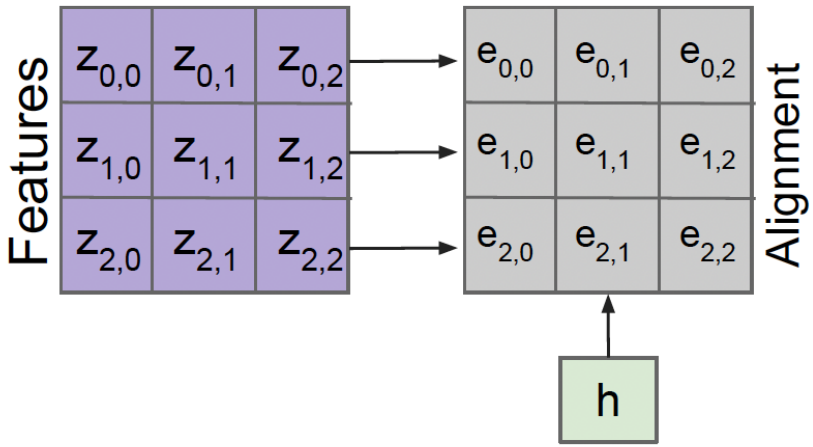
Inputs:
Features: **z** (shape: H x W x D)
Query: **h** (shape: D)

https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

General attention layer



Operations:
Alignment: $e_{i,j} = f_{att}(h, z_{i,j})$



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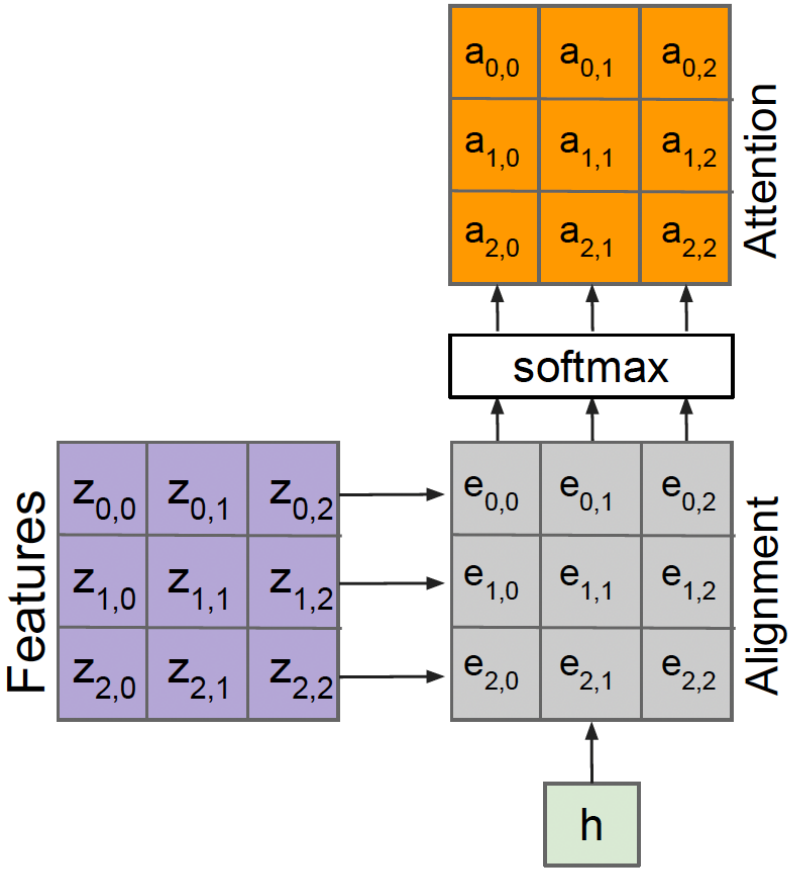
General attention layer



Operations:
Alignment: $e_{i,j} = f_{att}(h, z_{i,j})$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$

Inputs:
Features: **z** (shape: H x W x D)
Query: **h** (shape: D)

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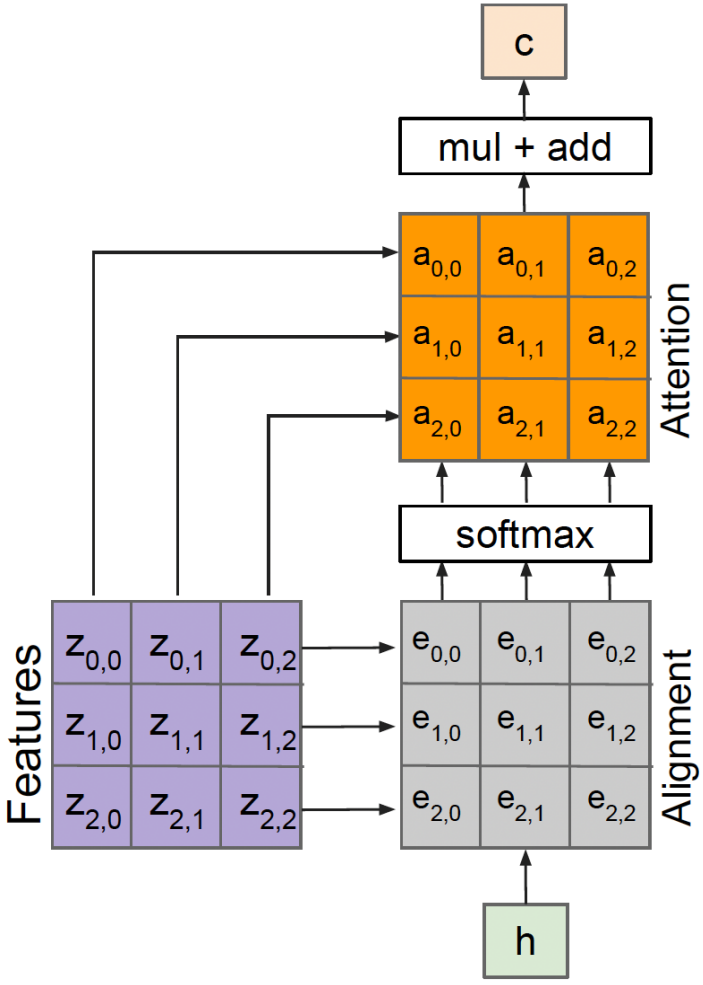
General attention layer



Outputs:
context vector: **c** (shape: D)

Operations:
Alignment: $e_{i,j} = f_{att}(h, z_{i,j})$
Attention: **a** = softmax(**e**)
Output: **c** = $\sum_{i,j} a_{i,j} z_{i,j}$

Inputs:
Features: **z** (shape: H x W x D)
Query: **h** (shape: D)

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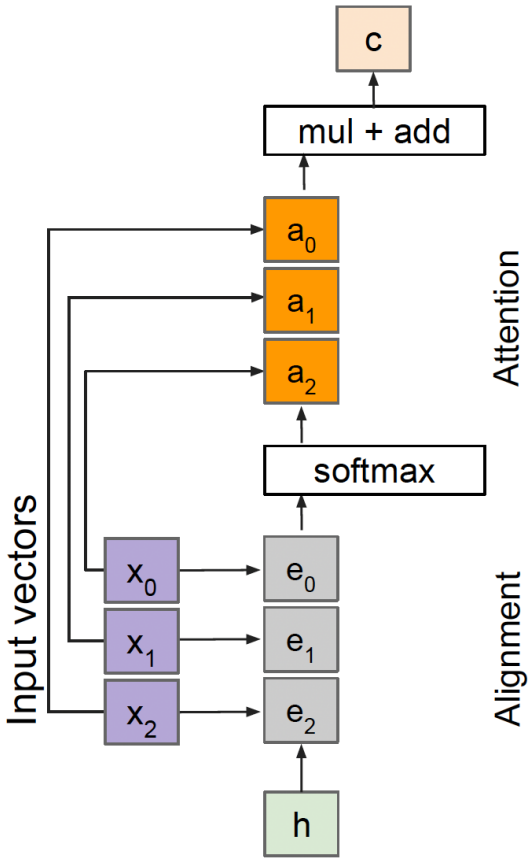
General attention layer



Outputs:
context vector: **c** (shape: D)

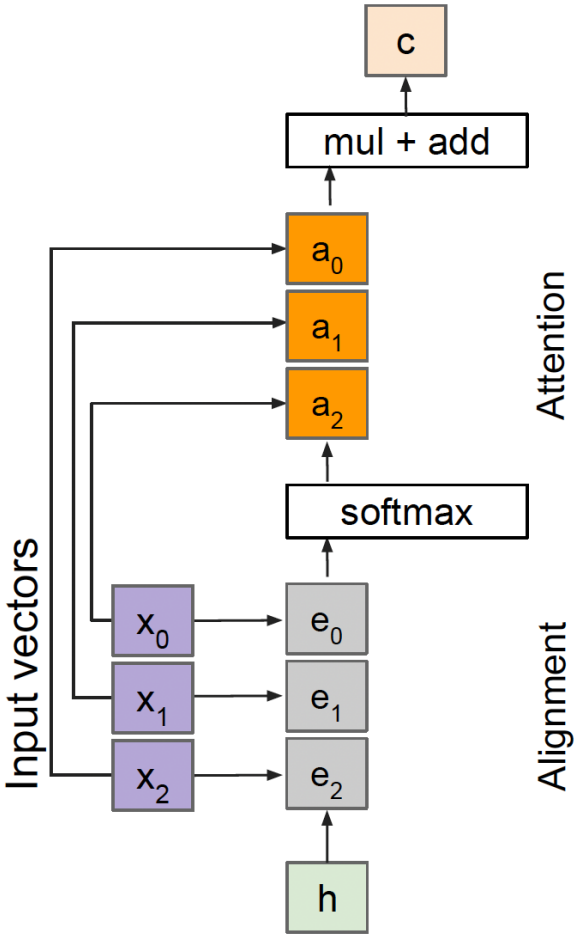
Operations:
Alignment: $e_i = f_{att}(h, x_i)$
Attention: **a** = softmax(**e**)
Output: **c** = $\sum_i a_i x_i$

Inputs:
Input vectors: **x** (shape: N x D)
Query: **h** (shape: D)

Attention operation is **permutation invariant.**
- Doesn't care about ordering of the features
- Stretch H x W = N into N vectors

https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

General attention layer



Outputs:
context vector: **c** (shape: D)

Operations:
Alignment: $e_i = h \cdot x_i$
Attention: **a** = softmax(**e**)
Output: **c** = $\sum_i a_i x_i$

Inputs:
Input vectors: **x** (shape: N x D)
Query: **h** (shape: D)

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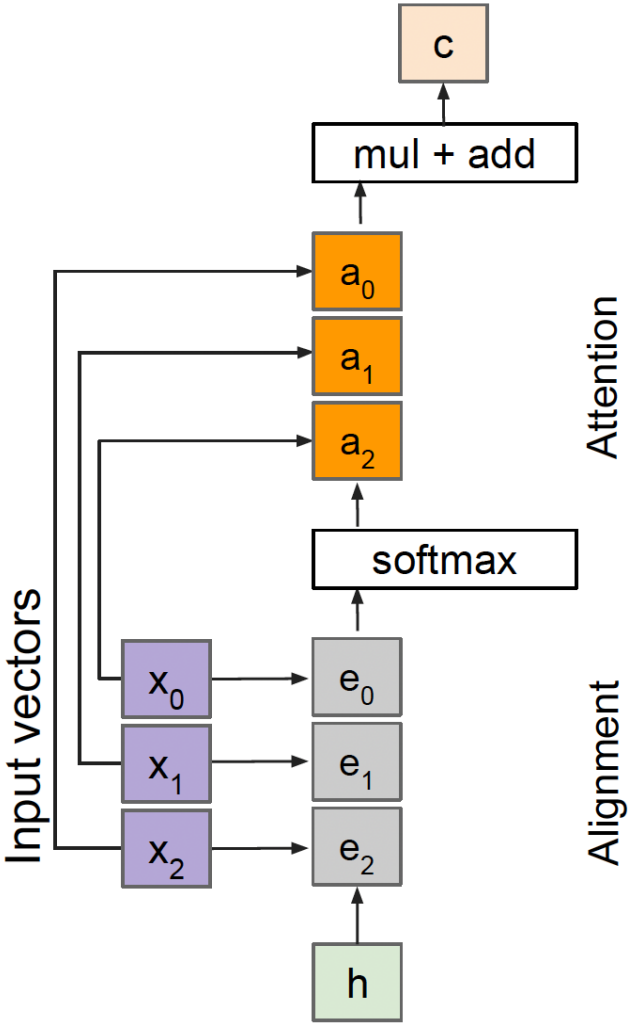
General attention layer



Outputs:
context vector: **c** (shape: D)

Operations:
Alignment: $e_i = h \cdot x_i / \sqrt{D}$
Attention: **a** = softmax(**e**)
Output: **c** = $\sum_i a_i x_i$

Inputs:
Input vectors: **x** (shape: N x D)
Query: **h** (shape: D)

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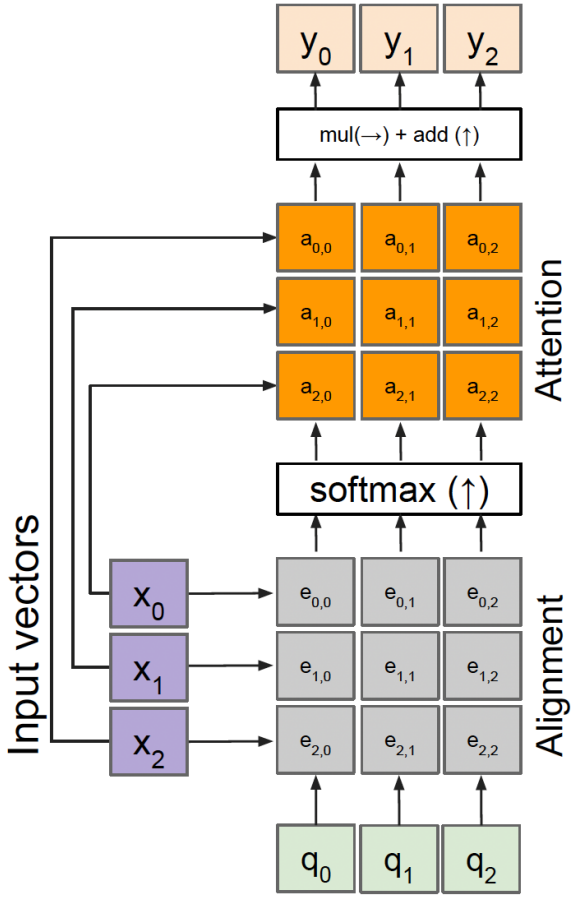
General attention layer



Outputs:
context vectors: **y** (shape: D)

Operations:
Alignment: $e_{i,j} = q_j \cdot x_i / \sqrt{D}$
Attention: **a** = softmax(**e**)
Output: $y_j = \sum_i a_{i,j} x_i$

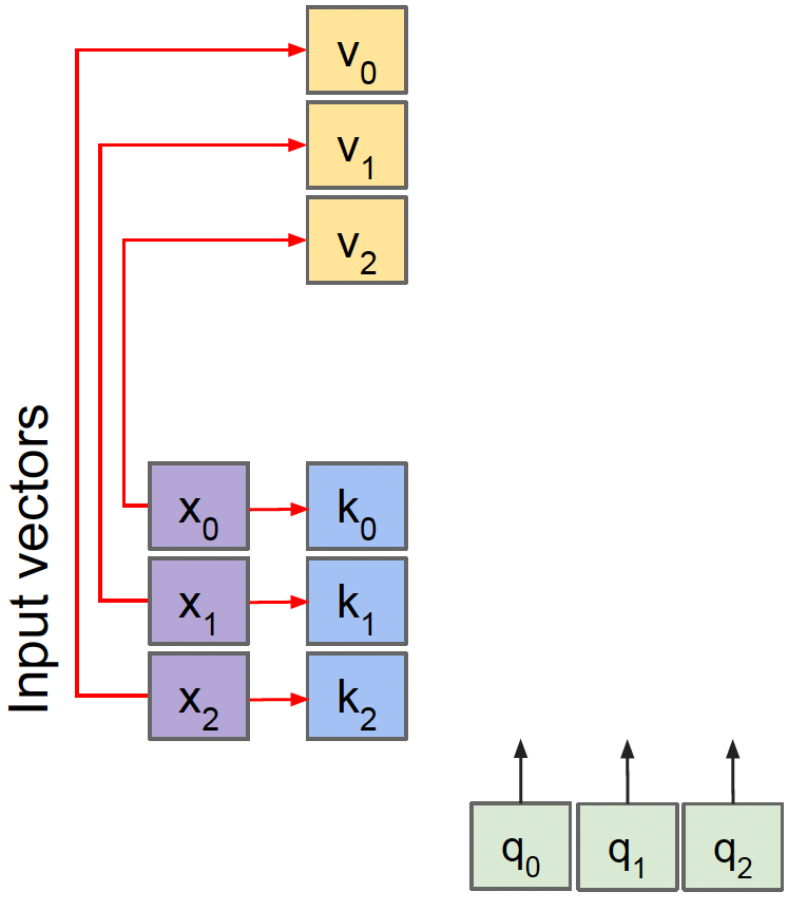
Inputs:
Input vectors: **x** (shape: N x D)
Queries: **q** (shape: M x D)

Multiple query vectors
- each query creates a new output context vector

Multiple query vectors

https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

General attention layer



Operations:
Key vectors: $k = xW_k$
Value vectors: $v = xW_v$

Inputs:
Input vectors: **x** (shape: N x D)
Queries: **q** (shape: M x D_k)

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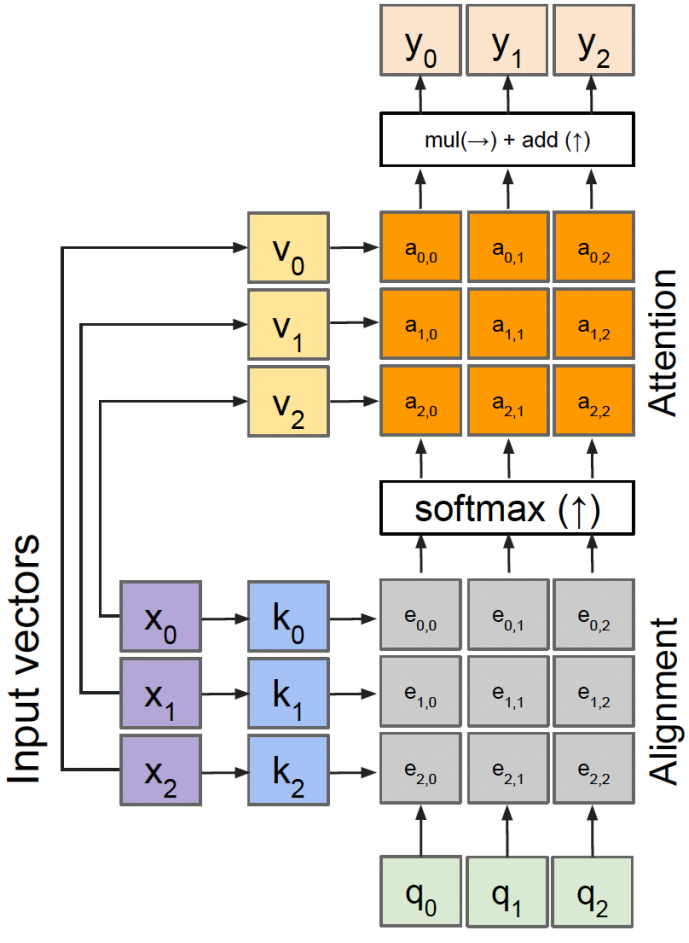
General attention layer



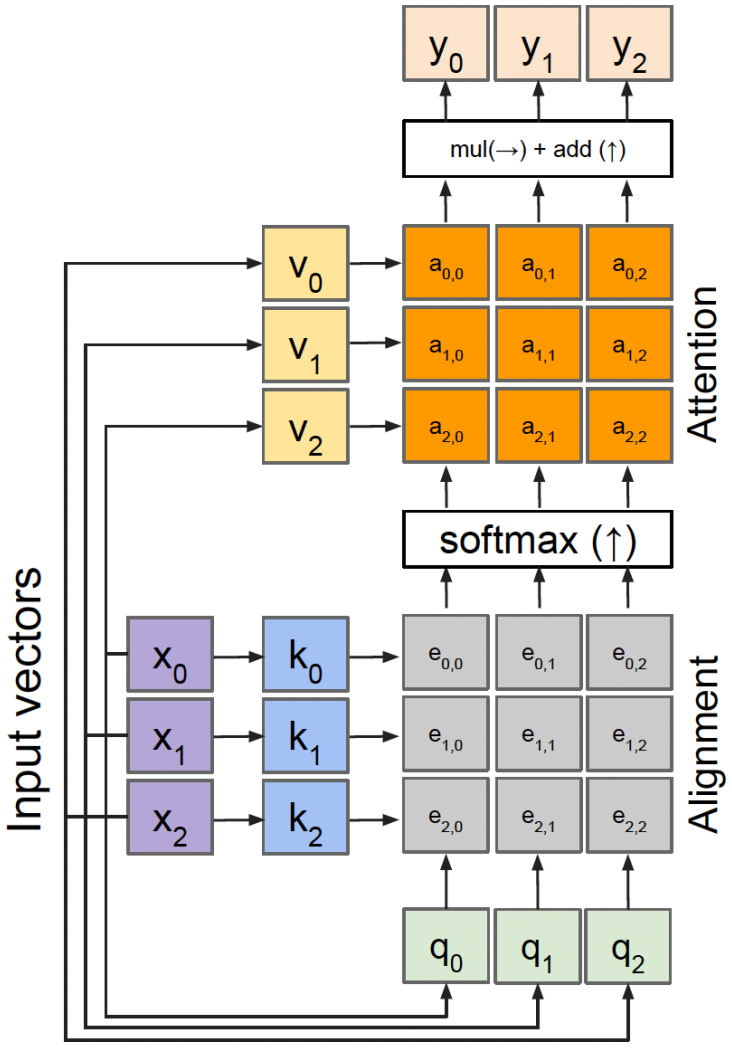
Outputs:
context vectors: **y** (shape: D_v)

Operations:
Key vectors: $\mathbf{k} = \mathbf{x}W_k$
Value vectors: $\mathbf{v} = \mathbf{x}W_v$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $y_j = \sum_i a_{i,j} v_i$

Inputs:
Input vectors: **x** (shape: N x D)
Queries: **q** (shape: M x D_k)

https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

Self attention mechanism



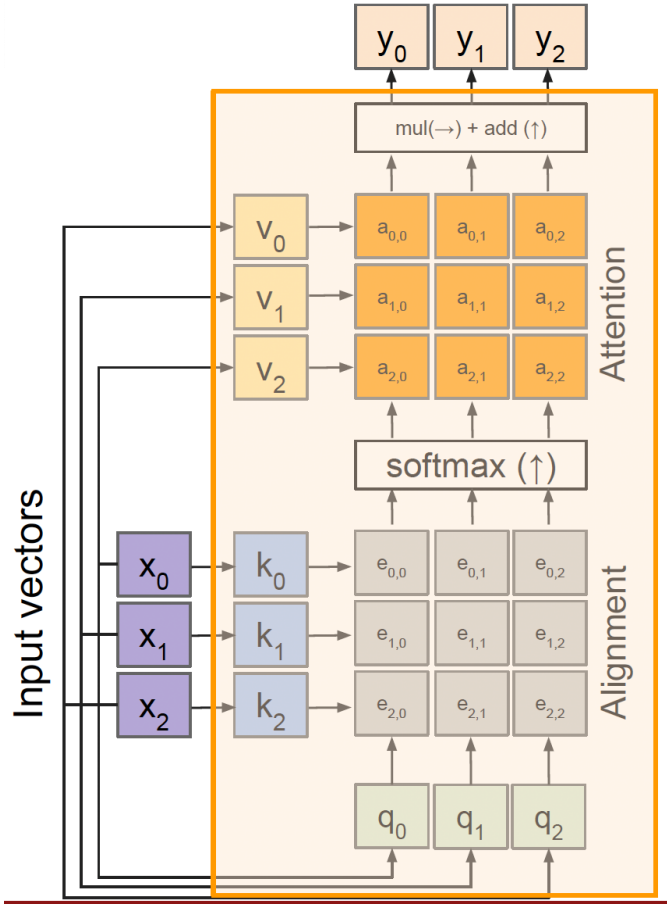
Outputs:
context vectors: **y** (shape: D_v)

Operations:
Key vectors: $\mathbf{k} = \mathbf{x}W_k$
Value vectors: $\mathbf{v} = \mathbf{x}W_v$
Query vectors: $\mathbf{q} = \mathbf{x}W_q$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $y_j = \sum_i a_{i,j} v_i$

Inputs:
Input vectors: **x** (shape: N x D)

https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

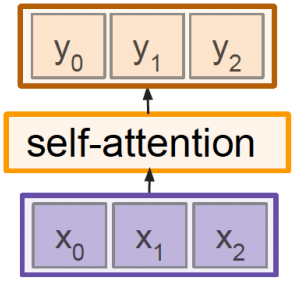
Self attention mechanism



Outputs:
context vectors: **y** (shape: D_v)

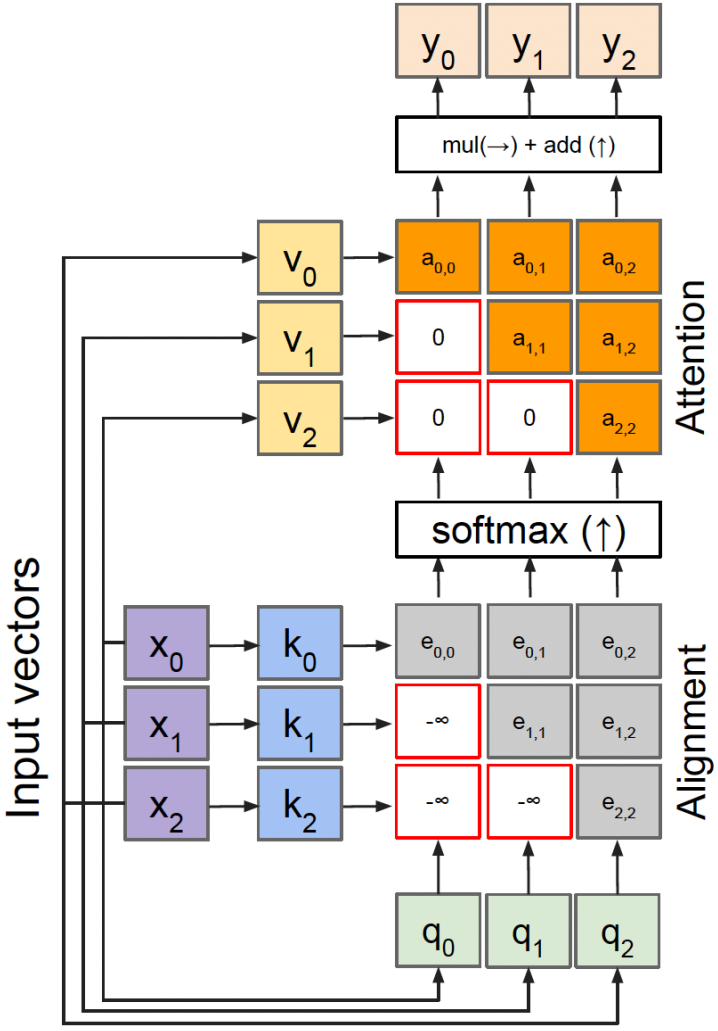
Operations:
Key vectors: $\mathbf{k} = \mathbf{x}W_k$
Value vectors: $\mathbf{v} = \mathbf{x}W_v$
Query vectors: $\mathbf{q} = \mathbf{x}W_q$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $y_j = \sum_i a_{i,j} v_i$

Inputs:
Input vectors: **x** (shape: N x D)



https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

Masked Self attention mechanism



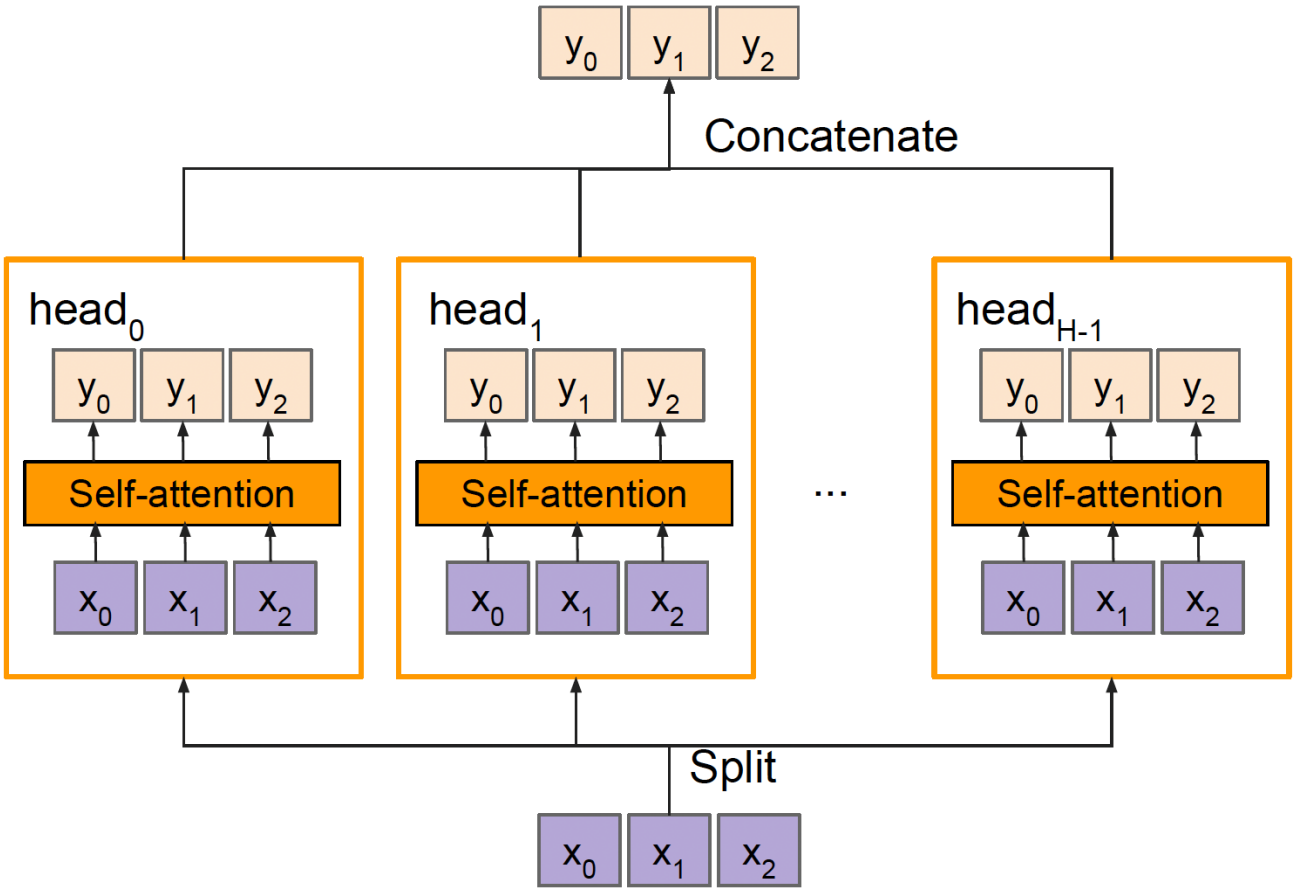
Outputs:
context vectors: **y** (shape: D_v)

Operations:
Key vectors: $\mathbf{k} = \mathbf{x}W_k$
Value vectors: $\mathbf{v} = \mathbf{x}W_v$
Query vectors: $\mathbf{q} = \mathbf{x}W_q$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $\mathbf{a} = \text{softmax}(\mathbf{e})$
Output: $y_j = \sum_i a_{i,j} v_i$

Inputs:
Input vectors: **x** (shape: N x D)

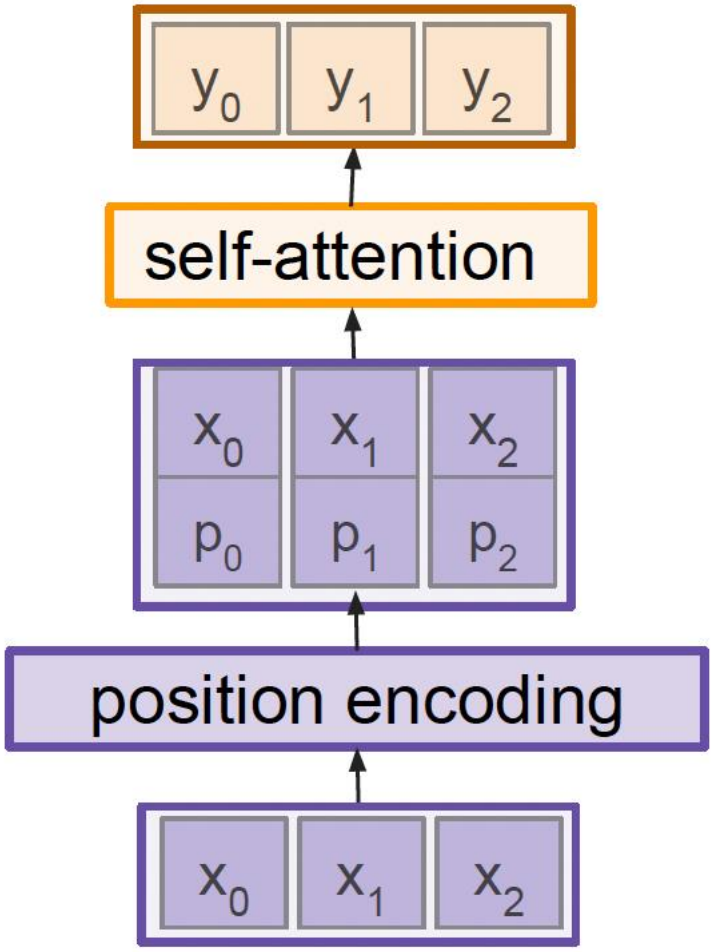
https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

Multi-head attention mechanism



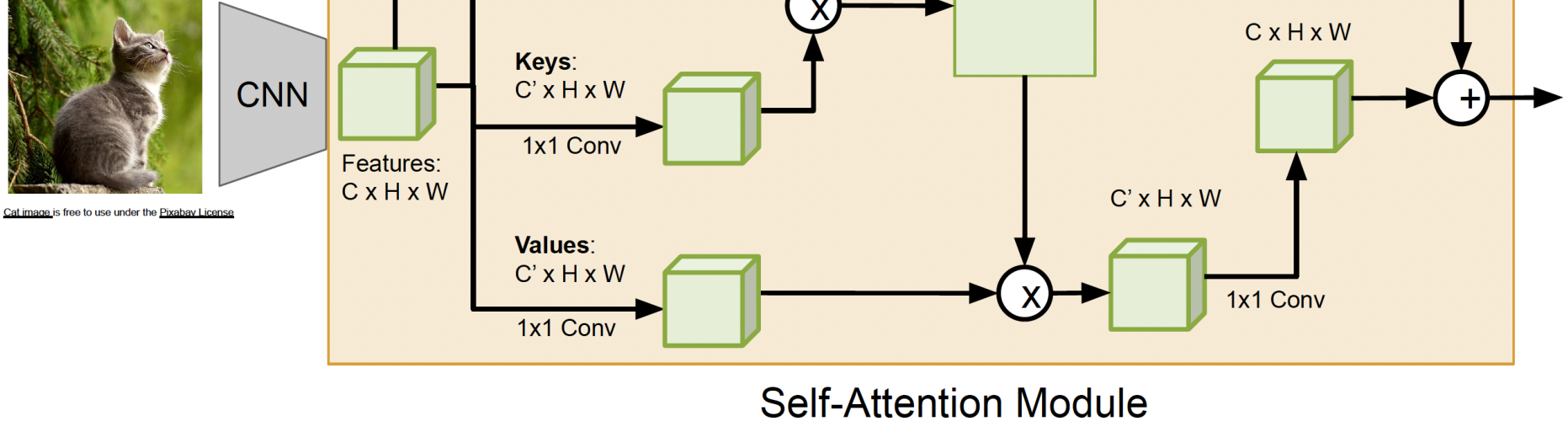
https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

Positional encoding



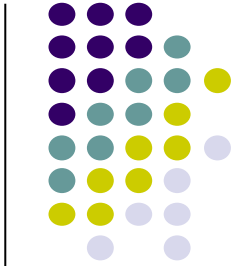
Vaswani et al, "Attention is all you need", NeurIPS 2017

CNN example with self attention layer



https://cs231n.stanford.edu/slides/2022/lecture_11_ruohan.pdf

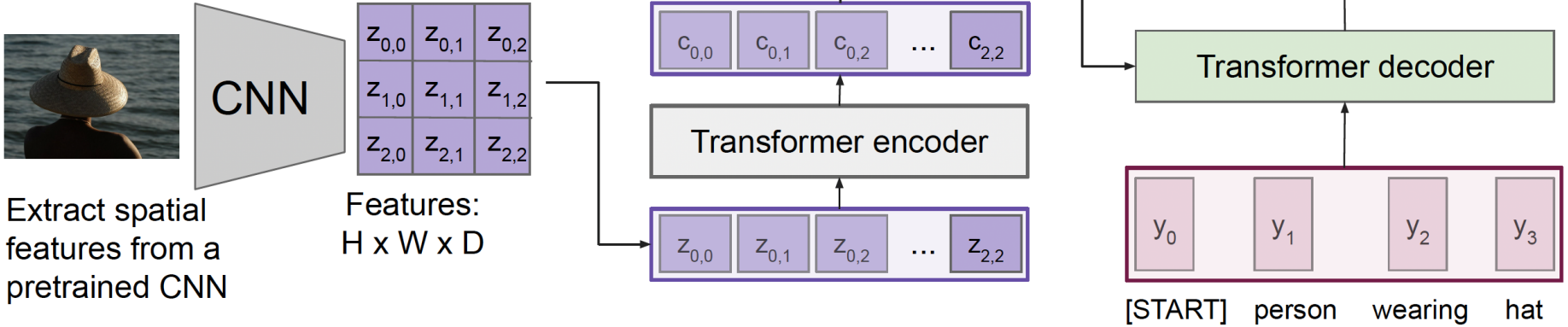
Transformer



Input: Image **I**
Output: Sequence **y** = $y_1, y_2, ..., y_T$

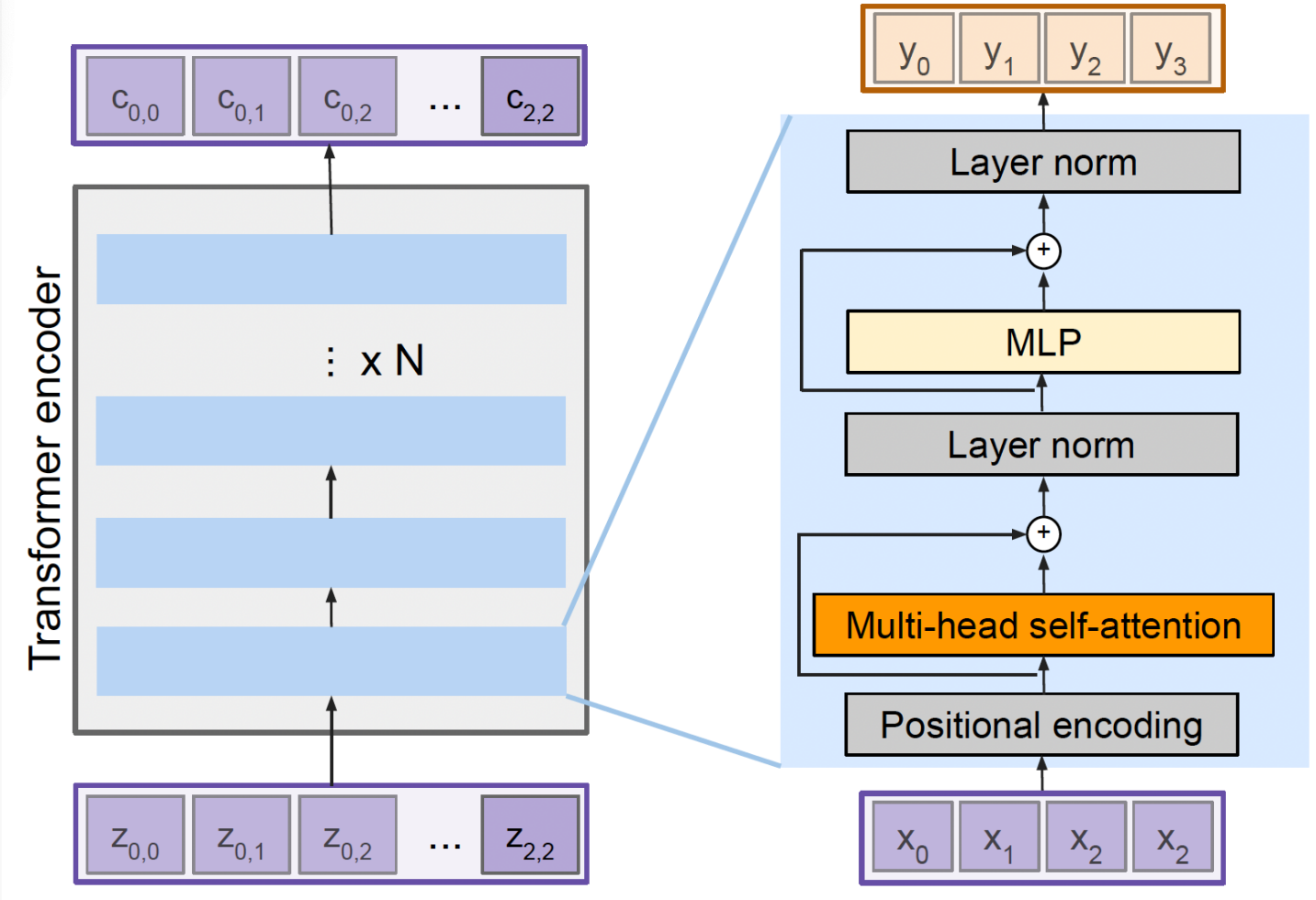
Encoder: $\mathbf{c} = T_W(\mathbf{z})$
where **z** is spatial CNN features
$T_W(.)$ is the transformer encoder

Decoder: $y_t = T_D(\mathbf{y}_{0:t-1}, \mathbf{c})$
where $T_D(.)$ is the transformer decoder



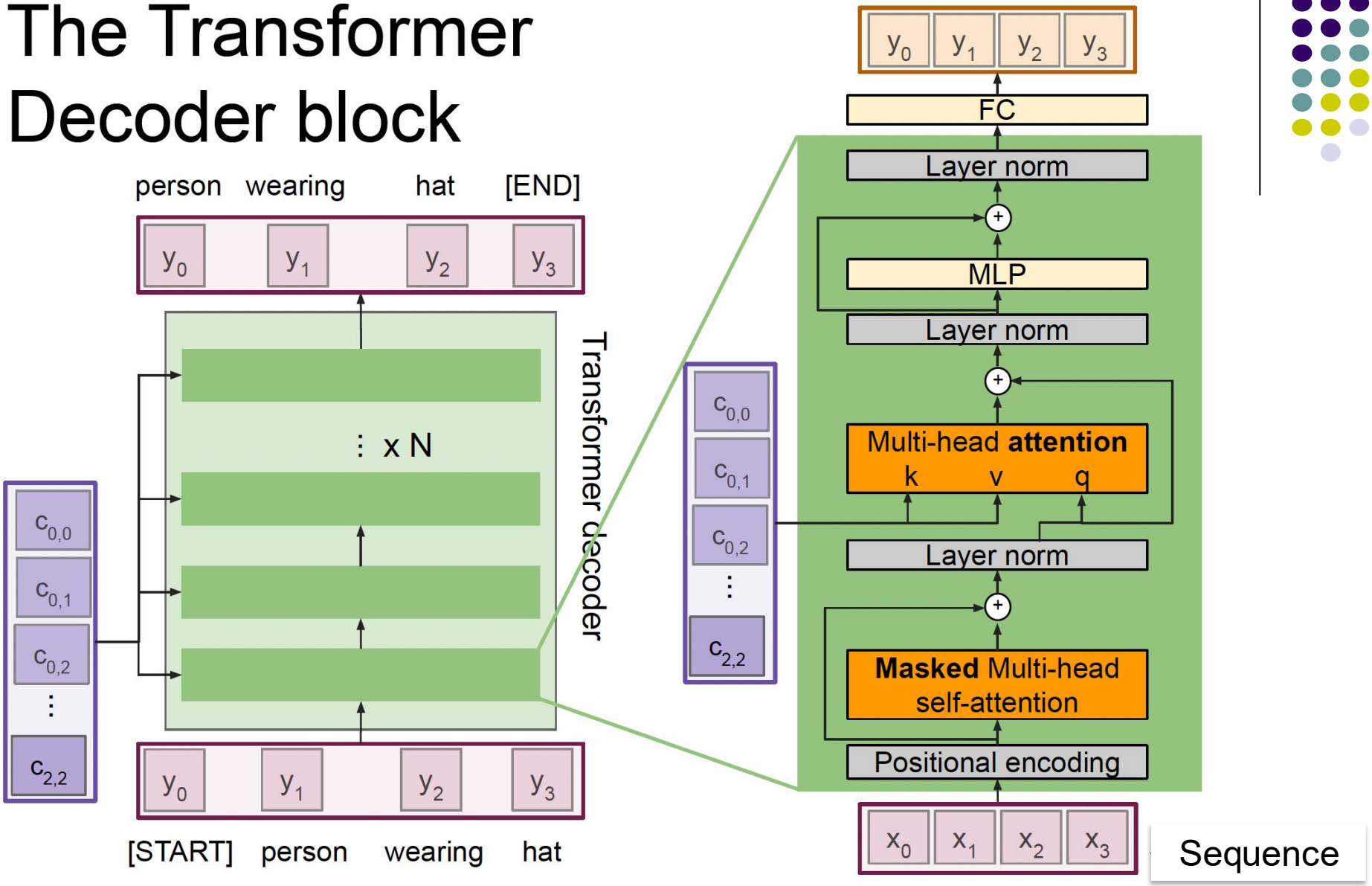
Vaswani et al, "Attention is all you need", NeurIPS 2017

Transformer encoder



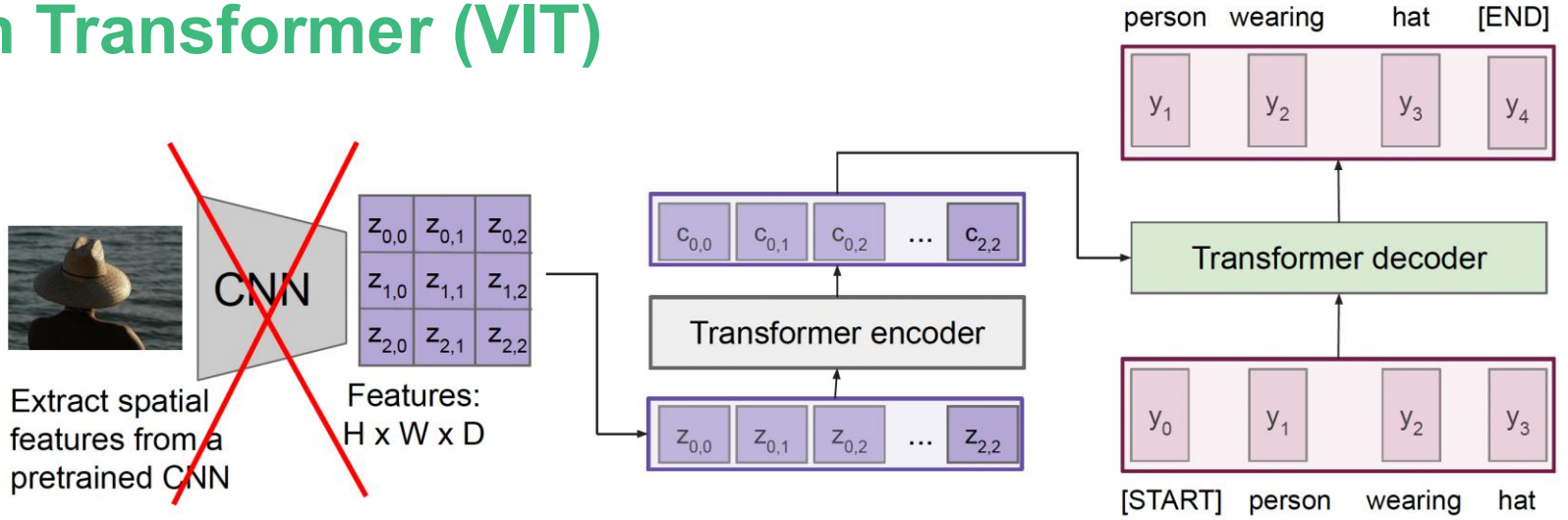
Vaswani et al, "Attention is all you need", NeurIPS 2017

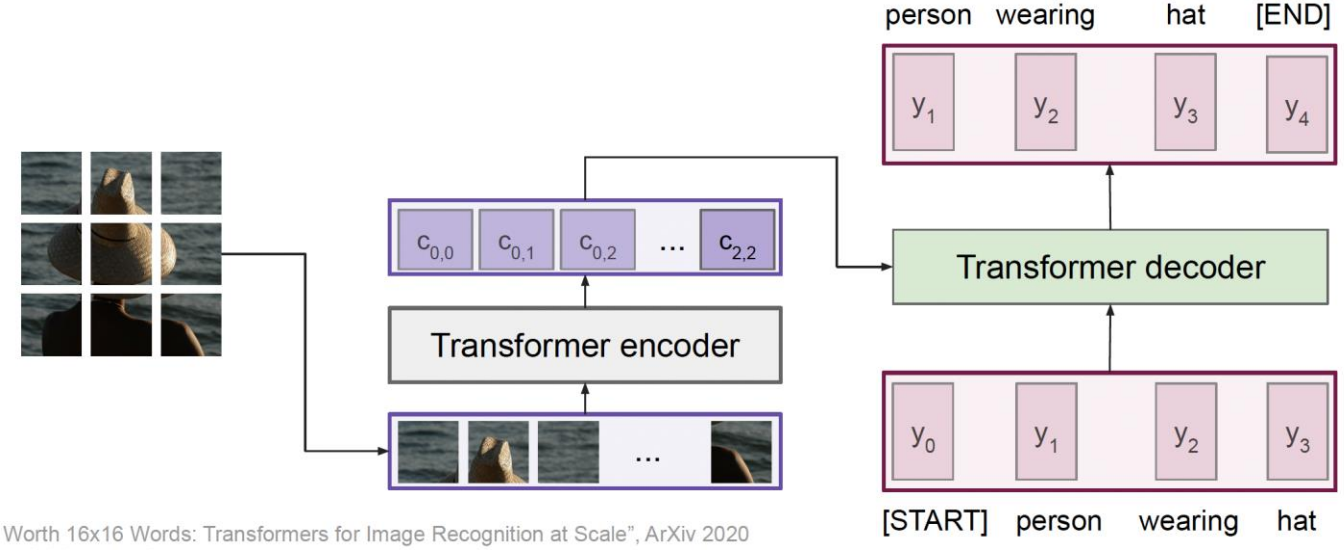
The Transformer Decoder block



Vaswani et al, "Attention is all you need", NeurIPS 2017

Vision Transformer (VIT)





Dosovitskiy et al, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale", ArXiv 2020

Colab link to an implementation of vision transformers